

Computational Interestingness

ERC Advanced Grant Submission

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1. Why is interestingness an interesting problem?

The entertainment and media industry produce an ever increasing amount of so-called digital *content*. These content differ substantially from traditional information or data processed by computers in that the affective dimension users endow them with is essential. Most of our modern industries – the *psycho technologies* according to (Stiegler, 2008) – movies, music, literature and media in general attempt to produce content that attract users in a way or another. As such, the question of understanding why people like or dislike specific objects, is at the root of modern economy, as illustrated by the emerging notions of *attention economy* (Davenport & Beck, 2002) or *emotional design* (Norman, 2004).

There are many reasons to understand what makes object likeable, or interesting: attention deficits disorders increase with the widespread use of television and internet, especially in young children (Christakis et al., 2004) as watchers are encouraged to have a passive attitude, and this becomes a major problem of our societies. New cognitive modes of attention are being created as a consequence of the passive exposure to huge quantities of content (Hayles, 2004), but there is a danger in letting these modes be uniquely determined by the evolution of the media industry, notably of video games.

Paradoxically however, there are still few theories and frameworks for understanding what « liking » means, from a computational perspective. The standard way of modeling the relation between humans and objects is through the notion of *taste*. Taste is traditionally considered as a *dictionary* associated to each user, that defines categories of preferred objects or features thereof. The main problem of the digital economy is then to match content with taste. However, there are many reasons to think that this vision is overly simplistic, because taste is not a consistent, static phenomenon, and to think that the relation between humans and cultural artefacts is much more complex.

The goal of our research is precisely to try to understand the processes that make things “interesting” in general. This goal is not directed by a mercantile perspective (i.e. how to produce likeable content), but by a user-centered vision of content: We would like to contribute to the design of systems in which people can not only navigate intuitively in huge content repositories, but create *themselves* objects of interest. This vision that the future of content resides in the ability of individuals to produce cultural artefacts as much as to access them was developed in (Pachet, 2008). To do so, we aim at developing a scientific approach to taste, departing from the vision of taste as a dictionary, but considering taste as the result of a complex interaction process between humans and digital objects, with a particular emphasis on temporal objects. As such, a central theme of investigation of our research is the provocative question “*What is interesting?*”. This question is problematic as a substantial body of research aims precisely at showing that what makes people like or express subjective judgments is essentially determined by social pressure (see the case of music preference as demonstrated by Salganik et al., 2006). However, if these works do show the importance of social pressure in the expression of preference, they also show that there is no uniform distribution of tastes among people: there are things that are objectively appealing, or interesting, at least to some categories of people, but the reasons for this attraction remain basically unknown. We refer to this problem as the “popularity feature problem”: What are the *features* of objects that can make them interesting, for a particular user, or category of users?

More generally, the question “what is interesting” relates to several domains: the descriptive side of *features* (signal processing), *attention* (psychology), *taste* (marketing), *well-being* (social science), and to some extent *beauty* (aesthetics).

These related questions represent an important emerging domain of investigation for computer science. We think indeed that interestingness is the “third main layer of communication” in increasing order of sophistication, after syntax (local arrangement of symbols) and meaning (neutral interpretation) and that at such it deserves an explicit treatment. We propose an approach to study interestingness based on the explicitation and testing of a number of hypotheses, together with experiments involving the modeling of artifacts and experimentations of their use. The main objective of our project is not to derive a psychological theory of interestingness (although such a theory could be derived as a side effect), but to investigate and develop software technologies that can assist users in *producing* interesting objects. To this aim, we think it is essential to study concurrently both the *perception* questions (i.e. what make things interesting?) and the *creation* aspect (how people create interesting objects? And which software technologies can help them do so?).

The project draws on research conducted in the last 10 years or so around this theme (see Section XX), and the realization that to make substantial progress requires a combination of techniques and expertise that exceeds what a single researcher can do. In particular, I propose to investigate the problem under three interrelated scientific perspectives:

- A *Computer Science* perspective, in which the major problem is the design of software technologies that are able to *model interestingness* of digital objects (notably temporal) in a descriptive way, and 2) produce, or assist in the production of interesting objects.
- A *Mathematical* perspective, in which several problems identified in the process of developing interactive systems are abstracted and considered. Notably novel techniques for understanding the topology of feature space, as well as the development of *feature generation* techniques as we have already initiated.
- An *Experimental* perspective, in which creative acts are studied both in natural settings (music composition, *story telling*) and in artificial contexts, i.e. through the interaction with an artificial system, notably what I have called *reflexive interactions*.

1.1 Existing works and theories of interestingness

In economics, the notion of preference has been largely studied under the umbrella of the utility theory and “Preference theory”. These models assume that users exhibit relatively consistent behaviors, which can be studied to derive their preferences, leading to a number of well-defined problems (such as the multi-criteria profit maximization problems). Recently *well-being* studies (Kahneman, 2004) emerged as a field in itself, borrowing from economics and the utility theory, while importing models of decision-making and the psychology of judgment.

In computer science, *collaborative filtering* (Shardanand & Maes, 1995) paved the way for building various categories of recommendation systems. In this view, preferences are modeled through the observation of social patterns: content is basically ignored. Recently, researchers in information retrieval have attempted to bridge the gap between purely social theories of taste (collaborative filtering) and content analysis in order to increase the accuracy of automatic recommendation systems (see the Netflix prize for instance). However these models are always based on naïve views of user taste, and low-level modeling of digital objects.

Information theory has recently been re-investigated in order to yield models of *surprise* or *curiosity*. The « Wundt curve » as predicted one century ago describes the typical experience of arousal as being optimal when achieving a compromise between repetition/boredom and surprise (Cf Figure 1). The Wundt curve predicts the typical inverted “U-shapes” of interestingness models based on information theory (random or unpredictable on the sides, optimally surprising in the middle). A naïve but appealing model of “computational beauty” was proposed by (Schmidhuber, 1997), based on the idea that a beautiful object is one that is “compressible”, and that interestingness is basically the first derivative of such a notion of beauty. Many developments were initiated along this line, e.g. in robotics with the notion of *artificial curiosity* (Kaplan & Oudeyer 2007), in artificial vision (Itti & Baldi 2005) or music analysis (Adballah & Plumbley 2008). All these models are based on the systematic comparison of a temporal predictor, possibly continuously trained, with reality as the

temporal object unfolds. Because these models are based on a domain independent model of reality (information), they remain at a level of generality which makes them interesting (...) but not falsifiable, in the Popperian sense. We can draw inspiration from them (we consider these models as relevant), but they do not answer our questions convincingly. Conversely, the signal processing community becomes increasingly aware and interested in analyzing human expressions, notably in the field of social signal processing (Vinciarelli et al., 2009). However, the range of problems are still low-level (e.g. emotion recognition through face analysis), and the question of interestingness is considered only implicitly.

The field of neuroesthetics is devoted to the question “what is beauty?” from a neuroscientific perspective (Kawabata & Zeki, 2004). This question is related to ours, but not equivalent, as we wish to remain independent of any aesthetic theory or value judgment: if the concept of beauty is doomed to remain elusive, interestingness can be measured by observing the behavior of subject, for instance the time of attention, as in we did in the Continuator project (Addessi & Pachet, 2005). These studies often consider beauty or appeal from a n “naturalistic” viewpoint, i.e. looking for general properties of nature as driving forces. A computational counterpart of neuroesthetics is the recent field of “computational aesthetics” (Hoenig, 2005) which attempts to model works of art using general, physics-inspired models of the world, such as fractals, or symmetry (Stewart, 2009). However, these tools are difficult to use in a content production context, and serve mostly as operational ways of defining or categorizing beauty.

In psychology, *experimental aesthetics*, pioneered by (Berlyne, 1974) is devoted to the understanding of what makes people feel beauty from a psychological perspective. This field led to spectacular developments such as the “introspection illusion” and “choice blindness” (Johansson et al., 2005), which reveal that the expression of subjective judgments is non intuitive and non trivial. Aesthetic preference has been compared to various general measures such as prototypicality (Whitfield, 1983). These ideas have applications to design, notably to help conduct user studies and so-called usability evaluations (Norman, 2004) but remain mostly at a metaphorical level which makes them unadapted for our purpose.

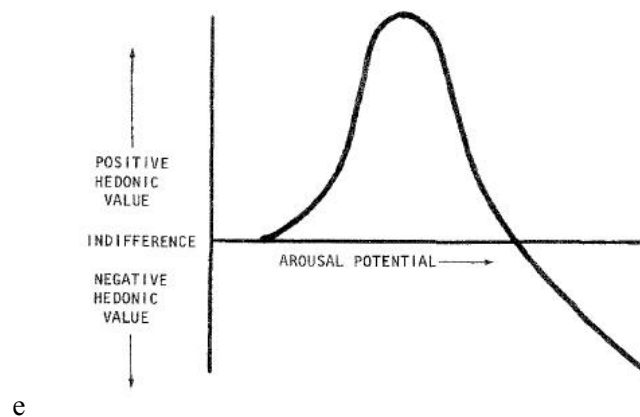


Figure 1. The Whundt curve, modeling the optimal arousal experience in an information-theoretic context

1.2 Conclusion

- 1) There is a growing but implicit recognition of the importance of interestingness as a field of study in many scientific domains,
- 2) There is diversified research in psychology and human studies in this area, although these works are not yet unified and recognized as belonging to a field *per se*,

- 3) Much of the current research is descriptive, i.e. concerns the problem of describing interesting objects or situations, or predicting user behavior. Little research is devoted to the study of the creation process, and its mediation through interactive systems.
- 4) Existing computational approaches are based on general hypothesis (e.g. information theory). These are not easy to validate, nor to use in production contexts.

We propose to investigate an *integrated computational approach to interestingness*, based on the following stances:

- 1) We study both the perception and the creation processes together. This will be done by carrying a number of experiments concerning the creation process. In particular we want to produce:
 - The first complete record of all the steps implied in the creation of a musical song, from scratch, in order to collect and analyse data. See Section “**Democratic composition**” where we will study the processes of decision-making in a song composition setting.
 - We want to carry experiments with birds (see Section **Artificial Ethology**) to validate hypothesis concerning reflexive interactions, as discussed in Section XX and already initiated with the LERCC laboratory.
- 2) Concerning the *features of interestingness*, we propose to explore other hypothesis than the ones considered so far, which are too general, and not sufficiently based on experimental evidence.
 - The “interesting = rare” hypothesis. This hypothesis implies that the perception of an object implicitly creates a combinatorial space in which the object is considered as a “rare solution”. Technically such a model implies the possibility of inferring a problem from the object to create, and using properties of solutions of this problem during the design process.
 - See Section **Rapid exploration of Markovian spaces**, and the Ryoan-ji stone garden example.
 - Experiments related to “what makes a solution interesting” in the field of 4-voice harmonization, see Section **Automatic Harmonization**
 - The “interesting = difficult to produce” hypothesis, as exemplified with the facial expression analysis observed in users manipulating artificial generation systems. See section **Virtuosity studies**.
 - The « interesting = myself » hypothesis. Interesting objects are interesting to the extent that they reveal hidden facets of the user, or help him/her achieve goals. The mimetic dimension of reflexive interactions as observed in the Continuator project. See Section **Reflexive interactions**.
 - The “interesting = a chaotic process” (i.e. mere exposition to artefacts change taste). The idea of using quantum theory insights for modeling decision-making is developed in (Mugur-Schächter et al., 2002). This analogy is to be taken as a metaphor, but we will carry some basic experiments concerning the relation between creation and preferences on simple musical objects (see Section **Features of Perception/Action**).

2. Project description

The project consists in studying the hypothesis expressed above, with the goal of achieving a unified, grounded computational theory of “active” interestingness. More precisely I want to study concurrently the following problems, listed as sub-projects that will be addressed concurrently.

2.1 Interactive composition and Hit Song Science

This project will focus on the study of the relations that exist between the “features of perception” and “features of Action”. The main intuition here is that the features of our preferred objects (say, melodies) are not exactly the features of the objects we tend to create naturally. This discrepancy is precisely what makes the creation act so difficult, in particular for simple objects. This project is split in several sub-projects.

2.1.1 Democratic composition

Democratic composition is devoted to the direct study of “what makes the composition act so difficult”. We will design and develop a compositional prototype, to study the compositional process during the elaboration of a complete musical piece. This will notably include all the “trial and error” schemes concerning the melodic, harmonic and timbral aspect of the piece. What is expected here is to highlight the topology of the search space which underlie the creation of “appealing” objects (here, melodies). The project will be carried in collaboration with a known pop song composer (Benoit Carré, from Lilicub, author of several hits, is a possibility). The goal is to produce in fine fully-fledged songs, up to their actual production and distribution, in order to gather also data about their “success”. One way to distribute the songs will be to participate in the *International Songwriting Competition*, held each year (<http://www.songwritingcompetition.com/>) organized by a panel of world-renowned artists.

The main idea of this prototype is to show that the decision-making process concerning song composition is a “singular path”: best solutions are often the simplest, but paradoxically the most difficult to find, and require a strong capacity to distance oneself from the object of creation. Measurements will be carried not only about the actual musical objects, but also about the context (moment of actions, possibly video recording of the situations, etc.)

Other questions: How consistent are subjective preferences for musical objects? Quantum nature of subjective evaluation.

Ph. D. Thesis #5 and Ph. D. Thesis #6.

Existing works: (Pachet, 1999), (Pachet, 2006), (Pachet & Roy, 2008), Preliminary interactive applets developed with Amaury (see demos).

2.1.2 Automatic Harmonization

This project will focus on the study of the technical features of interestingness, in a specific setting: harmonized melodies. We already proposed efficient techniques for computing all “legal” solutions of 4-voice Baroque harmonizations in a purely syntactic approach (Pachet & Roy, 2001). The techniques we proposed (based on finite-domain constraint satisfaction) can be used to compute quickly the set of all solutions, but do not allow to differentiate the “good” solutions from the other ones. Many works have been done since this paper on automatic harmonizations, but none has yet addressed the specific problem of distinguishing musically interesting solutions from the others. We will extend these studies to understand now how singular solutions can be identified from the set of legal ones, by using notably feature generation techniques (Pachet & Roy, 2009). Feature generation is a promising technique that can identify precisely unknown features, given well defined supervised classification tasks. We will also study harmonization in other domains than Baroque music (Jazz harmonization, in collaboration with the arranger Antoine Dessein). What is expected from this project is the identification of melodic-harmonic features, notably long-term features that can explain, at least partially, what is a “good” harmonization.

This work relates to our works on “hit song science” (Pachet & Roy, 2008). In this study we showed that current feature sets (in the audio domain) are not sufficiently precise to explain popularity. Here, we can address this problem in a limited context (symbolic, and style-specific), and try to make a contribution in this domain.

2.1.3 Features of Popularity

This work will focus on the signal processing approaches to music analysis. The main question is again, what are the features of interestingness, taken from an audio signal perspective. Starting from our negative result using general-purpose features (Pachet & Roy, 2008), we will investigate the use of feature generation techniques (Pachet & Roy, 2009b), which have already shown promising results for a number of complex audio classification tasks. However, these techniques are not directly applicable to the question of popularity because the analysis of full songs requires a statistical reduction (the bag of frame approach) which is not suited to the principles of feature generation. An extension of audio feature generation to handle “long” audio sequences is therefore necessary.

Another work we want to address is the incorporation of feature generation in information-theoretic approaches to music listening. Notably, we want to study how feature generation can be used to extend the works on information rate (e.g. Abdallah & Plumbley, 2009) which use primate audio features to handle more sophisticated examples.

Collaboration with Mirko Degli Esposti (Department of Mathematics, U. Bologna). Ph. D. Thesis #1.

Existing works: (Pachet & Roy, 2009), (Barbieri et al., 2009)

2. Models of Virtuosity

In this project, I want to study formal models able to account for *virtuoso generative behaviors*. More precisely, I have already pinpointed the “control” problem inherent to Markovian modeling, and proposed to see it as a combinatorial exploration problem (Pachet & Roy, 2009). This problem turns out to be very combinatorial indeed and I want to find heuristics or novel generation algorithms based on the observation of human behaviors.

2.2.1 Difficultuosity

In this project, I want to study the (hypothesized) correlation between combinatorial features and observable signs of interest. For instance, I would like to map specific facial expressions in music listening, Jazz in particular to the concept of “difficultuosity”. Difficultuosity can be defined as the objective technical difficulty the listener or user would experience if he/she were to produce the object he/she is listening to. Preliminary studies show that there is indeed such a link (see videos with Geber).

Collaboration with Goldsmith University (Mark d’Inverno, G. Wiggins). Ph. D. Thesis #3

Existing works: (Pachet, 2009)

2.2.2 Description-based design

Establish a formal link between machine-learning and combinatorial exploration. Insert here the description of description-based design. Find an analytical solution to the DBD problem.

Collaboration with J.D. Zucker (IRD). Ph. D. Thesis #2

Existing works: (Pachet, 2009)

2.2.3 Doodling Systems

Line sketching, to complete.

2.3 Artificial Ethology

In this project, I want to study the various hypothesis raised here in the context of animal interaction. The first motivation for such a study is to understand to which extent the basic principles of interestingness are human specific or more general. The second motivation is to the pioneering

experiments of Kreutzer et al. concerning the study of virtuosity in bird songs (Draganoiu et al. 2002). These authors showed that artificially synthesized canary songs optimizing specific “virtuosity features” substantially outperform natural songs.

We will extend these studies through the use of automated closed-loop reflexive systems, as experienced in (Lerch et al. 2009). Study preference systems in animals using “active ethology”. Artificial Ethology is a concept we have started to investigate based on the idea of provoking situations in animals rather than to rely on passive observation.

Collaboration with M. Kreutzer and L. Nagle, LECC. Ph. D. Thesis #4

Existing works: (Molnar et al., 2008), (Giret et al., 2008), (Lerch et al., 2009)

3. Workplan

5 years, 2 ME = 10 PhDs plus administrative staff, travels, equipment, collaborations.

4. Bio and self-promotion

My background is in computer science (Object-Oriented Programming) and Artificial Intelligence (Phd. on the integration of production rules and constraints in an Object oriented setting) with a long interest in music modeling. In 1998, I joined Sony CSL where I set up a research team devoted to music interaction. I was the inventor of pioneering music interaction technologies including: *playlist generation, dynamic mixing, audio feature generation, reflexive interaction, description-based design, interactive style imitation* and more recently *active ethology*.

Several technologies are being transferred within Sony (audio feature generation in software libraries for automatic classification of music and images, interactive music learning for music video games).

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